House Price Prediction — Summary of Findings

Internship Week 1 — Dataset: Kaggle Housing Prices Dataset (545 properties, 13 features)

Dataset & Approach

The dataset contains 545 house listings with area, room counts, amenities (air conditioning, hot water heating, parking, etc.), and furnishing status. After confirming there were zero missing values and zero duplicate rows, the six yes/no columns were mapped to 1/0 and furnishing status was one-hot encoded. A Linear Regression model and a Random Forest Regressor were trained on an 80/20 train/test split and compared.

Model Performance

Metric	Linear Regression	Random Forest
MAE (avg. error)	₹970,043	₹1,022,545
RMSE	₹1,324,507	₹1,403,779
R^2 (test set)	0.653	0.610
R^2 (5-fold CV, shuffled)	0.632	0.628

Both models explain roughly 60–65% of the variation in price, with a typical prediction error around ₹1 million on houses averaging about ₹4.8 million. Linear Regression edged out Random Forest slightly — with only 545 rows and largely linear relationships, the extra flexibility of a Random Forest wasn't an advantage here.

What Drives Price Most

- Area (square footage) — the single strongest driver in both models.
- Number of bathrooms — a close second, and one of the most “actionable” features for renovation.
- Air conditioning, number of stories, and parking — meaningful but secondary.
- Furnishing status and bedroom count had comparatively little independent effect once area and bathrooms were accounted for.

Surprises in the Data

- The raw CSV is sorted by price (highest to lowest). This is harmless for train_test_split (which shuffles by default) but produced wildly negative cross-validation scores when run with an un-shuffled KFold — a reminder to always check sort order before cross-validating.
- Zero missing values across all 545 rows is unusual for “real-world” data, suggesting this is a cleaned teaching dataset rather than raw scraped listings.

Recommendation for a Real Estate Business

Since area and bathroom count are the strongest, most controllable price levers, pre-sale renovation budgets should prioritize adding usable space or an extra bathroom over cosmetic upgrades like furnishing — this is likely to move sale price more per rupee invested.